A Multimodal Adversarial Attack Method via Frequency Domain Enhancement and Fine-Grained Cross-Modal Guidance

Yaguan Qian, Qinqin Yu, Qiqi Bao, Shouling Ji, *Member, IEEE*, Wei Wang, *Member, IEEE*, Bin Wang, Zhaoquan Gu, *Member, IEEE*, and Zhen Lei, *Fellow, IEEE*

***Abstract*—Vision-language pretraining (VLP) models have demonstrated outstanding performance in image-text understanding tasks but remain highly susceptible to transferable adversarial attacks. While ensemble-based guided attacks improve adversarial transferability by increasing the diversity of image-text pairs, they primarily rely on spatial-domain data augmentation, which can lead to model overfitting to image details and limit the generalization capability of attacks. To address this limitation, this study proposes a frequency-domain adjustment-based adversarial attack method that modifies specific frequency components of input images to reduce detail interference and enhance the stability of adversarial examples. Additionally, a fine-grained feature extraction technique is introduced to optimize image-text alignment, further improving the transferability of cross-modal attacks. Experimental results demonstrate that the proposed method achieves superior attack transferability and generalization performance across two major VLP architectures, fusion models and alignment models, as well as multiple tasks on the Flickr30 K and MSCOCO datasets.**

Received 12 April 2025; revised 8 July 2025; accepted 16 August 2025. Date of publication 21 August 2025; date of current version 4 November 2025. This work was supported in part by the National Key Research and Development Program of China under Grant 2022YFB3102100, in part by National Natural Science Foundation of China under Grant 62476250, Grant 62472335, Grant 62372137, Grant 62172243, Grant 62402425, Grant 62402418, Grant U23B2054, Grant U244120033, and Grant U24A20336, in part by Zhejiang Provincial Natural Science Foundation of China under Grant LQN25F010018 and Grant LD24F020002, in part by the "Pioneer and Leading Goose" R&D Program of Zhejiang under Grant 2025C01082 and Grant 2025C02263, in part by the Shenzhen Science and Technology Program under Grant KJZD20240903103811016, and in part by Postgraduate Research and Innovation Foundation of Zhejiang University of Science and Technology under Grant 2024yjskc19. *(Corresponding authors: Qiqi Bao; Zhaoquan Gu.)*

Yaguan Qian, Qinqin Yu, and Qiqi Bao are with the School of Artificial Intelligence and Information Engineering, Zhejiang University of Science and Technology, Hangzhou 310023, China (e-mail: qianyaguan@zust.edu.cn; 222309252021@zust.edu.cn; bqq2024@zust.edu.cn).

Shouling Ji is with the College of Computer Science and Technology, Zhejiang University, Hangzhou 310058, China (e-mail: sji@zju.edu.cn).

Wei Wang is with the Ministry of Education Key Lab for Intelligent Networks and Network Security, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: wei.wang@xjtu.edu.cn).

Bin Wang is with the Zhejiang Key Laboratory of Artificial Intelligence of Things (AIoT) Network and Data Security, Hangzhou 310052, China (e-mail: wangbin2@hikvision.com).

Zhaoquan Gu is with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), Shenzhen 518055, China (e-mail: guzhaoquan@hit.edu.cn).

Zhen Lei is with the Center for Biometrics and Security Research and National Laboratory of Pattern Recognition, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China (e-mail: zlei@nlpr.ia.ac.cn).

This article has supplementary downloadable material available at https://doi.org/10.1109/TDSC.2025.3601232, provided by the authors.

Digital Object Identifier 10.1109/TDSC.2025.3601232

***Index Terms*—Multimodal attack, adversarial transferability, frequency domain enhancement, cross-modal guidance.**

I. Introduction

RECENT advancements in VLP models have significantly improved multimodal tasks such as Image-Text Retrieval (ITR) [1], [2], Visual Grounding (VG) [3], [4], [5], and Image Captioning (IC) [6], [7], [8] through large-scale image-text pretraining. Concurrently, the adversarial robustness of VLP models has gained increasing attention, prompting the development of various attack strategies to enhance black-box attack effectiveness by improving the transferability of adversarial examples.

However, existing adversarial attack methods predominantly rely on spatial-domain augmentation techniques (e.g., rotation, cropping, flipping) to diversify adversarial examples [9], [10], [11], [12]. These methods, which directly manipulate the original data, often cause models to overfit to image details, thereby limiting the generalization ability of adversarial perturbations. Moreover, studies [13], [14], [15] have shown that key object positions in images tend to be relatively fixed, leading different models to focus on similar spatial regions. In contrast, models may exhibit varying sensitivity to different frequency components [16], [17], such as high-frequency and low-frequency information—a factor largely overlooked by traditional spatial-domain attack strategies [10], [12], [18], [19]. This suggests that optimizing adversarial example generation from a frequency-domain perspective presents a promising alternative.

Specifically, in the image modality, high-frequency perturbations are highly vulnerable to smoothing operations (such as Gaussian blur), which reduces the robustness of adversarial examples. Conversely, low-frequency perturbations have been shown to exert a more stable influence on model predictions [20]. In the text modality, word-level features can be analogous to high-frequency components in images, reflecting detailed information, while sentence-level features resemble low-frequency components, capturing global semantics. However, most existing adversarial attack research [10], [12], [18], [19] focus primarily on perturbing word-level features, overlooking the fine-grained mapping relationships across different feature levels. This limitation constrains the optimization of the text modality and hinders the generation of effective cross-modal adversarial examples.

1545-5971 © 2025 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission. See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

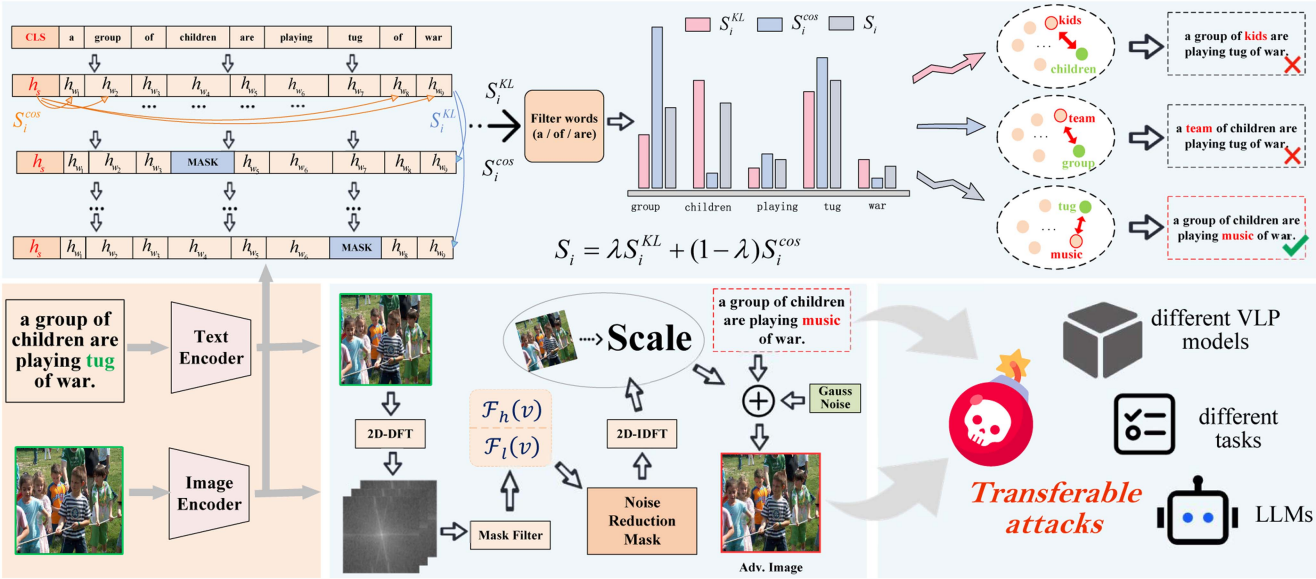


Fig. 1. Overview of our proposed attack method. The upper part illustrates the generation process of adversarial texts, highlighting the advantages of our multi-level feature extraction strategy over baseline methods. "KL" denotes the Kullback-Leibler divergence loss term (7), and "cos" denotes the cosine similarity loss term (8). These terms are combined as a weighted sum with coefficient λ (9), where $\lambda \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. The results under different λ settings are reported in Tables V and VI. The lower part demonstrates the introduction of a frequency-domain enhancement mechanism guided by adversarial texts to generate adversarial images, ultimately enabling attack transferability across different model architectures, tasks, and LLMs.

To address this, we propose a novel attack method that integrates both frequency-domain and spatial-domain transformations. In the image modality, the method selectively adjusts high- and low-frequency components to mitigate noise and detail interference. In the text modality, it employs a multi-level feature mapping strategy to fully capture the fine-grained relationships between word-level and sentence-level features. This joint optimization effectively enhances the transferability and effectiveness of cross-modal adversarial examples. The overall attack process is illustrated in Fig. 1.

We evaluate our proposed method on two different VLP model architectures [21] (i.e., fusion-based and alignment-based models) using two publicly available multimodal datasets, Flickr30K [22] and MSCOCO [23]. Experiments conducted on three vision-language downstream tasks—ITR, IC, and VG—demonstrate that our method consistently generates highly transferable multimodal adversarial examples. Furthermore, when applied to large language models (LLMs), adversarial examples generated from VLP models retain a certain degree of effectiveness, further highlighting the robustness of our approach.

The main contributions of this paper can be summarized as follows:

- From a frequency-domain perspective, we introduce a frequency-domain enhancement mechanism in the image modality. By modifying certain frequency components of the input, we effectively reduce detail and noise interference, making the generated adversarial examples more stable.
- Inspired by the frequency-domain analysis of images, we propose a multi-level feature extraction strategy for the text modality. By capturing fine-grained relationships between image and text features at different levels, our approach enhances cross-modal interaction guidance, thereby improving attack transferability.
- Extensive experiments demonstrate the superiority of our method over baseline approaches. In cross-model architecture and cross-task attacks, our method improves the attack success transfer rate by 16%-18% compared to baselines. Furthermore, adversarial examples generated from VLP models remain partially effective when transferred to LLMs, highlighting the robustness of our attack strategy.

II. Related Work

A. VLP Models and Typical Tasks

VLP models have achieved remarkable progress in multimodal learning by jointly modeling images and texts using large-scale image-text pairs. Architecturally, existing VLP models can be broadly divided into two types: fusion-based and alignment-based models [21]. These two paradigms reflect different strategies for bridging visual and textual modalities, each offering complementary advantages in representation learning and retrieval effectiveness. Fusion-based models, such as ALBEF [24] and TCL [25], encode visual and textual features separately and fuse them via cross-modal attention mechanisms. In contrast, alignment-based models, such as CLIP [26], project visual and textual features into a shared semantic space using contrastive learning objectives to achieve efficient multimodal alignment.

Typical downstream tasks of VLP models include ITR [1], [2], VG [3], [4], [5], and IC [6], [7], [8]. ITR consists of two sub-tasks: text retrieval (TR) and image retrieval (IR), and serves as a fundamental benchmark for assessing cross-modal alignment capabilities. VG focuses on locating image regions

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

corresponding to textual expressions, and IC aims to generate coherent textual descriptions based on visual inputs. In this work, we adopt ITR as the primary attack task and evaluate cross-task transferability of adversarial examples through VG and IC, enabling a comprehensive assessment of adversarial generalization in multimodal settings.

B. Adversarial Transferability Attack

Multimodal adversarial attacks have emerged as a key approach for evaluating and improving the robustness of VLP models. Existing attack methods can be categorized into three types: image-only attacks, text-only attacks, and joint image-text attacks.

Projected Gradient Descent (PGD) [27] is a classic white-box image attack that perturbs pixel-level representations via gradient-based optimization, but it is limited to the visual modality. BERT-Attack [18] targets the text modality by replacing key words to mislead language models, without considering interactions with visual features. Sep-Attack applies independent perturbations to image and text modalities, representing an early attempt at multimodal adversarial learning, yet it lacks a coordinated interaction mechanism. To address the limitations of separate perturbations, Co-Attack [19] proposed a white-box strategy that jointly perturbs visual and textual inputs to disrupt cross-modal alignment in VLP models. Building on this idea, SGA [10] introduced spatial-domain data augmentation by constructing multi-scale image-text pairs to improve the diversity and transferability of adversarial examples in black-box settings. SGA is one of the most representative and effective black-box multimodal attack methods to date. Building upon SGA, DRA [12] proposed a diversification strategy along the intersection region of the adversarial trajectory, further enhancing robustness and cross-model transferability. In parallel, TMM [28] introduced an attention-guided feature perturbation method that leverages the self-attention matrices of vision-language models to generate more transferable adversarial examples, particularly in black-box scenarios.

Despite their effectiveness, these approaches still suffer from two main limitations. First, they rely heavily on spatial-domain perturbations, which can lead to overfitting to image details and reduce generalization across models. Second, the interaction between modalities is often shallow, limiting the ability to break deep semantic alignments. To overcome these challenges, our method introduces a novel multimodal attack strategy that integrates a frequency-domain enhancement mechanism and a multi-level text-guided perturbation scheme. This approach optimizes the structural representation of images while disrupting fine-grained semantic alignments across modalities, thereby significantly improving cross-architecture and cross-task adversarial transferability.

C. Adversarial Examples in Frequency Domain

With the growing interest in understanding the frequency response of deep learning models, frequency-domain adversarial attacks have emerged as a promising direction to improve the stability and generalization of adversarial examples. Tsuzuku and Sato [29] first proposed a Fourier-based framework to analyze model sensitivity to different frequency components. Subsequently, Yin et al. [30] introduced perturbation analysis in the Fourier domain to further investigate model vulnerability to high- and low-frequency distortions.

A series of studies have demonstrated that low-frequency perturbations exhibit greater stability than high-frequency ones and are more robust to pre-processing operations such as Gaussian blurring and denoising. For instance, Guo et al. [31] showed that constraining perturbations to the low-frequency subspace can enhance the efficiency of boundary-based black-box attacks. Zhou et al. [32] found that suppressing high-frequency perturbations improves transferability, while Sharma et al. [20] reported that low-frequency constraints not only accelerate adversarial generation but also boost cross-model generalization. In contrast, applying perturbations to other frequency regions often leads to a drop in attack effectiveness. Recent analyses have further highlighted the dependence of deep models on low-frequency features. Specifically, Zhang et al. [33] demonstrated that low-frequency components are essential for capturing global structural information and ensuring the stability of learned representations, and that removing or perturbing these signals can substantially compromise the overall robustness of the models. In parallel, the findings of Kim et al. [34] indicate that certain models exhibit pronounced sensitivity to low-frequency and phase components, rendering them particularly vulnerable to low-frequency phase perturbations and more prone to representational degradation under such conditions.

Despite these advances, most existing frequency-based approaches focus solely on the visual modality and overlook the cross-modal interaction with language. To address this gap, we propose a novel adversarial example generation method that combines frequency-domain modulation with fine-grained image-text guidance, effectively enhancing the generalization performance of multimodal attacks across tasks and architectures.

III. Methodology

A. Background and Motivation

Existing adversarial attack methods typically employ spatial-domain augmentations to enhance the transferability of adversarial examples in black-box models. Techniques such as rotation, cropping, and flipping increase sample diversity. However, these augmentation techniques remain dependent on transformations of the original data, which can lead to model overfitting to details and noise during training, ultimately limiting the generalization ability of adversarial examples.

Previous studies have demonstrated that different models rely on distinct frequency components in their decision-making processes. However, traditional approaches often neglect this factor, focusing exclusively on perturbation optimization in the spatial domain. To address this limitation, we adopt a frequency-domain perspective, utilizing augmentation mechanisms that modify specific frequency components of the input. This approach effectively mitigates the influence of details and noise, enhancing the stability of adversarial examples.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

Furthermore, in cross-modal tasks, word-level and sentence-level features in the textual modality can be analogous to high-frequency and low-frequency components in images. However, existing methods primarily focus on word-level optimization, neglecting the fine-grained mapping relationships between different feature levels. This limitation constrains the generation capability of cross-modal adversarial examples. To address this issue, we propose a method that integrates Discrete Fourier Transform (DFT) with a refined cross-modal guidance mechanism to optimize adversarial example generation from a frequency-domain perspective. By enhancing low-frequency information in images, optimizing multi-level features in the textual modality, and incorporating spatial-domain augmentation techniques, our approach improves the robustness and transferability of adversarial examples.

B. Dual-Domain Hierarchical Augmentation Strategy

1) Frequency-Domain Augmentation Mechanism: The core idea of the frequency-domain augmentation mechanism is to incorporate frequency-domain information to refine the essential features of an image while eliminating redundant high-frequency components. This process enhances the conciseness and stability of image representations. The frequency components of an image can be categorized into low-frequency and high-frequency elements, where the low-frequency components generally contain global structural information, while the high-frequency components carry details and noise.

First, we utilize the DFT to decompose a signal into spectral components of varying frequencies and magnitudes, enabling the modification of specific frequency components in the input. Given a two-dimensional image $v \in \mathbb{R}^{d\times d}$, its DFT representation is given by: $x = \mathcal{F}(v)$, where x represents the spectral components obtained through DFT, with values corresponding to the magnitudes of the basis functions. Smaller values indicate low-frequency components, while larger values correspond to high-frequency components. Since DFT is an invertible transformation, the original image can be reconstructed using the Inverse Discrete Fourier Transform (IDFT): $v = \mathcal{F}^{-1}(x)$.

To remove high-frequency components from the input image v, we apply a binary mask to its DFT transform $\mathcal{F}(v)$. Specifically, we introduce a binary mask matrix $M \in \{0,1\}^{d\times d}$ and perform element-wise multiplication:

$$x' = M \odot \mathcal{F}(v), \tag{1}$$

where $\odot$ denotes element-wise multiplication. The transformed image is then reconstructed using the IDFT:

$$\tilde{v} = \mathcal{F}^{-1}\left(M \odot \mathcal{F}(v)\right). \tag{2}$$

In our implementation, the binary frequency mask M is constructed by setting a square region of size $M \times M$ centered at the zero-frequency component in the DFT spectrum to 1, while all other elements are set to 0. After element-wise multiplication with the DFT of the input image, the masked frequency representation is transformed back to the spatial domain via IDFT to produce the filtered image that serves as the input to subsequent optimization steps.

This process yields a more concise and stable image representation by eliminating unnecessary details and noise. By reducing overfitting risks and enhancing essential image features, this frequency-domain augmentation improves the robustness of adversarial examples.

2) Spatial-Domain Augmentation Mechanism: Although frequency-domain augmentation can remove unnecessary noise and extract core features, we further incorporate spatial-domain augmentation to improve the transferability of adversarial examples. This approach expands the input space while maintaining cross-modal alignment consistency. Unlike traditional methods that generate adversarial examples based solely on individual image-text pairs, we constructs clusters of images and texts to enhance diversity.

For each image v, we select the most relevant textual pairs from the dataset to construct an augmented text set. Simultaneously, we apply multi-scale transformations and introduce Gaussian noise to the image v. The final expanded image set can be represented as: $V' = \{v'_1, v'_2, \ldots, v'_n\}, \quad T' = \{t'_1, t'_2, \ldots, t'_m\}$ where V' denotes the expanded image set after applying multi-scale transformations and noise perturbations, and T' represents the corresponding augmented text set. Adversarial examples are then generated using this enriched input set, enhancing their robustness and transferability.

C. Fine-Grained Image-Text Guidance Adversarial Attacks

In multimodal adversarial attacks, image-text interaction is essential for improving model transferability and cross-task performance. Traditional approaches typically rely on coarse-grained modality pairing information, which fails to fully capture the deep correlations between images and texts, thereby limiting the optimization effectiveness of adversarial examples.

Inspired by frequency-domain augmentation in images, this paper introduces a multi-level text feature extraction strategy to explore more fine-grained matching relationships between images and texts, enabling more precise interaction guidance. Compared to traditional methods, this strategy more effectively disrupts the associations between image and text modalities, leading to the generation of adversarial examples with higher transferability and robustness. This, in turn, provides stronger attack capabilities and better generalization for multimodal adversarial attacks.

1) Image-Guided Adversarial Text Generation: First, leveraging image embedding information, we employ a fine-grained text generation strategy to automatically select and replace the most influential words, generating adversarial texts that effectively disrupt the target model's decision-making process. This procedure consists of the following steps:

Feature Extraction: Feature extraction is fundamental to adversarial example generation, as it transforms raw text and images into high-dimensional representations with semantic or visual information. Inspired by the high- and low-frequency characteristics in the frequency domain of images, we employ the pretrained language model BERT for text feature extraction at both the word and sentence levels.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

For word-level feature extraction, each input text is first tokenized into a word sequence $w = \{w_1, w_2, \ldots, w_N\}$, and word feature representations h_w are generated as follows:

$$h_{w_i} = \mathcal{L}(w_i), \quad w_i \in t, \quad \forall i \in \{1, 2, \ldots, N\}, \tag{3}$$

where $\mathcal{L}$ represents the pretrained language model BERT [35], which converts each word into a high-dimensional vector h_{w_i}. This process yields a word-level feature set for the entire text.

For sentence feature extraction, the sentence representation is obtained by extracting the embedding of the [CLS] token. Given an input text t, after tokenization and encoding, the encoded sequence is represented as $t = [CLS, w_1, w_2, \ldots, w_N]$. The sentence-level feature representation is given by:

$$h_s = \mathcal{L}(\text{CLS}) = h_t[0], \tag{4}$$

where h_t represents the hidden state sequence output by the BERT, and $h_t[0]$ is the representation of the [CLS] token, serving as the feature representation of the entire sentence.

For image-guided adversarial text generation, we also need to extract image embedding features. The image v is transformed into an embedding representation through an image encoding network:

$$h_v = \mathcal{C}(V). \tag{5}$$

Generating Adversarial Text: To generate adversarial text, we first apply a masking process where each word w_i in the input text is replaced with the [MASK] token, forming a masked text set: $T_{mask} = \{t_1, t_2, \ldots, t_N\}$, where $t_i = t$ with w_i masked. Next, a text encoder $\mathcal{L}$ extracts embeddings for both the original text t and the masked text T_{mask}:

$$h_t = \mathcal{L}(t), \quad h_{mask,i} = \mathcal{L}(t_i), \quad \forall t_i \in T_{mask}. \tag{6}$$

To assess the impact of the word w_i on the entire sentence, we compute the Kullback-Leibler (KL) divergence between the original text embedding h_t and each masked text embedding $h_{mask,i}$:

$$S_i^{KL} = D_{KL}\left(\sigma(h_{mask,i}) \parallel \sigma(h_t)\right), \tag{7}$$

where $D_{KL}(\cdot||\cdot)$ represents the KL divergence, and $\sigma(\cdot)$ denotes the softmax operation. This score quantifies the degree of information change caused by the word w_i in the entire text, where a higher value indicates a more critical word.

Furthermore, we compute the cosine similarity between the word-level feature h_{w_i} and the sentence-level feature h_s to measure the contextual relevance of the word w_i:

$$S_i^{cos} = 1 - \frac{h_{w_i} \cdot h_s}{\|h_{w_i}\| \|h_s\|}. \tag{8}$$

A lower S_i^{cos} value suggests a stronger association between the word and the entire sentence, making it a more important target for adversarial attacks.

The final word importance score S_i is obtained by combining the KL divergence and the cosine similarity score using a weighted sum:

$$S_i = \lambda S_i^{KL} + (1 - \lambda) S_i^{cos}, \tag{9}$$

where λ is a hyperparameter controlling the contribution of KL divergence and cosine similarity to the final score.

After computing the importance scores for all words w_i, we sort them in descending order and select the top k words for substitution, thereby generating the adversarial text t^{adv}.

Iterative Optimization of Adversarial Text: To ensure that text perturbations can effectively influence the model, we use image embeddings as guidance to optimize the adversarial text embeddings in the gradient direction:

$$t_{i+1}^{adv} = t_i^{adv} + \alpha \cdot \text{sgn}\left(\frac{\sum_{j=1}^{\epsilon_t} \nabla h_{v_j}(h_v, h_t)}{\left\|\sum_{j=1}^{\epsilon_t} \nabla h_{v_j}(h_v, h_t)\right\|}\right), \tag{10}$$

where t_i^{adv} denotes the adversarial text at iteration i, α is the step size, and ϵ_t represents the number of perturbed tokens in the text and serves as the upper bound for text perturbation. $h_{v_j}(h_v, h_t)$ denotes the loss term computed for the j-th token, where h_v and h_t are the visual and textual embeddings, respectively. $\nabla h_{v_j}(h_v, h_t)$ denotes the gradient of the loss term with respect to the textual embedding. The sgn function normalizes the accumulated gradient over all tokens to stabilize updates.

2) Text-Guided Adversarial Image Generation: Next, based on adversarial text embedding information, we combine the image frequency-domain transformation strategy and multi-scale augmentation introduced in Section III-B to generate adversarial images. First, we compute the similarity matrix between the image embedding h_{v_i} and the adversarial text embedding h_{tj}^{adv}:

$$\text{sim}_{ij} = h_{v_i} \cdot h_{tj}^{adv^T}, \tag{11}$$

where h_{v_i} represents the embedding of the i-th image, and h_{tj}^{adv} is the embedding of the j-th adversarial text. Based on the similarity values, we generate the label matrix B_{ij}, where the elements indicate whether an adversarial text corresponds to an image, while all other entries are set to zero:

$$B_{ij} = \begin{cases} 1, & \text{if } i \text{ corresponds to } j \\ 0, & \text{otherwise} \end{cases}. \tag{12}$$

Then, we attack the images within the image-text matching space to maximize their deviation from the adversarial text set. To balance the matching degree of image-text alignment while ensuring that perturbations remain within a predefined range, the adversarial image is updated using a projected gradient descent step:

$$V_{t+1}^{adv} = \text{Clip}_{V \in \epsilon_i}\left(V_t^{adv} + \alpha \cdot \text{sgn}\left(\frac{\sum_{i,j} \nabla V_{t,adv}(\text{sim}_{ij} B_{ij})}{\left\|\sum_{i,j} \nabla V_{t,adv}(\text{sim}_{ij} B_{ij})\right\|}\right)\right), \tag{13}$$

where V_t^{adv} denotes the adversarial image at iteration t, α is the step size, and ϵ_i represents the maximum allowable perturbation for the image. $\text{sim}_{ij} B_{ij}$ denotes the similarity score between the i-th visual patch and the j-th textual token in the batch. $\nabla V_{t,adv}(\text{sim}_{ij} B_{ij})$ denotes the gradient of the similarity score with respect to the visual embedding. The Clip operator constrains the perturbation within the ϵ_i-ball defined by the L_p norm.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

The detailed description of the attack process can be found in Algorithm 1 of the supplemental materials.

IV. Experiments

A. Experimental Setup

1) Dataset: We evaluate our approach on the ITR task using two multimodal datasets: Flickr30K [22] and MSCOCO [23]. The Flickr30 K dataset consists of 31,783 images, each accompanied by five textual descriptions. Similarly, the MSCOCO dataset contains 123,287 images, with each image being annotated with an average of five captions. Additionally, we use the RefCOCO+ dataset [36] to evaluate the visual grounding task. This dataset comprises 19,992 images from MSCOCO, containing 50,000 target objects and providing 141,564 referring expressions. RefCOCO+ is designed to assess a model's ability to localize objects based on natural language descriptions. For the image captioning task, which is a vision-and-language problem, we conduct experiments and analysis using the MSCOCO dataset.

2) Models: We evaluate two common types of VLP models: fusion-based VLP and alignment-based VLP. For fusion-based VLP models, we select ALBEF [24] and TCL [25] as our research subjects. ALBEF consists of a 12-layer Vision Transformer (ViT-B/16) and two 6-layer Transformers, which are used for image encoding, text encoding, and multimodal feature fusion, respectively. TCL follows the same model architecture as ALBEF but adopts different pretraining objectives. For alignment-based VLP models, we evaluate CLIP [26]. CLIP provides two different choices for the image encoder, based on ViT-B/16 and ResNet-101, respectively, for image feature extraction.

3) Baselines: To verify the effectiveness of our proposed method, we compare it with various attack methods. These methods include PGD [27], BERT-Attack [18], Sep-Attack, Co-Attack [19], and SGA [10].

4) Parameter & Attack Settings: During the experiments, to ensure fairness, all methods adopt the same adversarial attack settings. For text modality attacks, we use BERT-Attack [18], where the perturbation range is set to $\epsilon_t = 1$, and the word candidate list length is set to $k = 10$. For image modality attacks, we use the PGD [27] method, where the perturbation range is set to $\epsilon_v = 8/255$, the step size is $\alpha = 2/255$, and the attack is executed for $T = 10$ iterations. Additionally, our image scaling strategy follows the method proposed by SGA [10], resizing the images to five different scales: $S = \{0.50, 0.75, 1.00, 1.25, 1.50\}$. When generating adversarial texts, we set the balancing parameter $\lambda = 0.7$. During the image frequency domain enhancement process, we set the mask range $M = 20$. The specific selection strategy can be found in Section IV-E.

5) Metrics: We use Attack Success Rate (ASR) as the evaluation metric to measure adversarial robustness and transferability in both white-box and black-box environments. Specifically, ASR reflects the proportion of successfully generated adversarial examples. A higher ASR indicates stronger transferability of the adversarial examples.

In addition, this study introduces the Structural Similarity Index (SSIM), Peak Signal-to-Noise Ratio (PSNR), and Fréchet Inception Distance (FID) as criteria for evaluating the quality of adversarial examples. SSIM measures the structural similarity between adversarial examples and the original images, ranging from 0 to 1. Values closer to 1 indicate higher structural fidelity. Here, SSIM > 0.8 is set as the threshold to ensure that perturbations do not significantly alter structural information, maintaining visual naturalness. PSNR assesses the level of noise introduced, measured in decibels (dB). Generally, PSNR above 30 dB ensures that image quality degradation is imperceptible to the human eye. Therefore, PSNR > 30 dB is adopted as the minimum quality requirement. FID quantifies the distribution discrepancy between generated and real samples in the Inception feature space, providing an accepted assessment of perceptual similarity and diversity. Lower FID values indicate higher fidelity, and scores below 50 are considered good visual quality.

B. Main Experiments

In this section, we systematically investigate the transferability of adversarial examples in fusion-based and alignment-based VLP model architectures using the Flickr30 K and MSCOCO datasets. We select four VLP models, including ALBEF and TCL, and evaluate them on the ITR task. Specifically, we generate adversarial examples constrained by the proposed method's parameter settings using one of the models and validate their effectiveness through a series of experiments. The evaluation includes one white-box attack and three different types of black-box attacks to comprehensively measure the transferability of adversarial examples.

To demonstrate the effectiveness of our method, we adopt the SGA [10] method as the baseline and compare it with the following attack strategies: PGD (an image-based attack) [27], BERT-Attack (a text-only attack) [18], Sep-Attack (an attack separately disturbing text and image), and Co-Attack (a modality-coordinated attack) [19]. Table I and Table II summarize the ASR on the TR R@1 and IR R@1 metrics for these methods on the Flickr30 K and MSCOCO datasets. More details on the experiments conducted on the Flickr30 K and MSCOCO datasets can be found in the Appendix I, available online, of the supplemental materials.

First, we evaluate the performance of different methods in a white-box attack setting. Experimental results show that our method achieves an ASR exceeding 99.9% across all target models in both the TR R@1 and IR R@1 metrics. While SGA already demonstrates strong attack performance in white-box environments, our improved approach still achieves slight but consistent improvements.

However, considering that adversarial attacks in real-world applications typically occur in black-box environments, we further focus on analyzing the transferability of adversarial examples. Given the architectural differences in VLP models, we divide the transferability analysis into two parts: attack effectiveness within the same model architecture and attack effectiveness across different model architectures.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

TABLE I
COMPARISON WITH MAINSTREAM METHODS IN ITR ON THE FLICKR30 K DATASET

Source	Attack	Ratio	FID	ALBEF		TCL		CLIP-ViT		CLIP-CNN	
				TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1
ALBEF	PGD	85.62	36.37	93.74	94.43	24.03	27.90	10.67	15.82	14.05	19.11
	BERT-Attack	—	—	11.57	27.46	12.64	28.07	29.33	43.17	32.69	46.11
	Sep-Attack	85.62	36.37	95.72	96.14	39.30	51.79	34.11	45.72	35.76	47.92
	Co-Attack	86.90	37.82	97.08	98.36	39.52	51.24	29.82	38.82	31.29	41.99
	SGA	**89.54**	34.73	99.90	**99.98**	87.88	88.05	36.69	46.78	39.59	49.78
	DRA	89.48	**34.32**	99.79	99.93	90.78	91.76	52.52	61.05	54.02	63.60
	Ours	89.40	34.56	**100.00**	99.93	**91.25**	**92.45**	**53.01**	**63.69**	**57.22**	**66.38**
TCL	PGD	81.18	43.34	35.77	41.67	99.37	99.33	10.18	16.30	14.81	21.10
	BERT-Attack	—	—	11.89	26.82	14.54	29.17	29.69	44.49	33.46	46.07
	Sep-Attack	81.18	43.34	52.45	61.44	99.58	99.45	37.06	45.81	37.42	49.91
	Co-Attack	82.53	40.47	49.84	60.36	91.68	95.48	32.64	42.69	32.06	47.82
	SGA	84.16	38.19	92.49	92.77	**100.00**	**100.00**	36.81	46.97	41.89	51.53
	DRA	83.58	38.80	93.58	**94.97**	**100.00**	99.98	51.78	61.02	56.83	65.59
	Ours	**84.30**	**37.02**	**93.85**	94.18	**100.00**	**100.00**	**54.23**	**64.59**	**59.26**	**68.71**
CLIP-ViT	PGD	78.48	45.07	3.13	6.48	4.43	8.83	69.33	84.79	13.03	17.43
	BERT-Attack	—	—	9.59	22.64	11.80	25.07	28.34	39.08	30.40	37.43
	Sep-Attack	78.48	45.07	7.61	20.58	10.12	20.74	76.93	87.44	29.89	38.32
	Co-Attack	77.31	43.67	49.84	20.18	10.01	21.29	78.53	87.50	29.50	38.49
	SGA	79.16	40.34	22.42	34.59	25.08	36.45	**100.00**	**100.00**	53.26	61.10
	DRA	79.36	41.32	**28.36**	43.06	**29.72**	44.05	**100.00**	**100.00**	64.64	69.40
	Ours	**80.28**	**40.05**	27.84	**43.22**	29.19	**45.26**	**100.00**	**100.00**	**65.13**	**73.48**
CLIP-CNN	PGD	81.61	42.87	2.29	6.15	4.53	8.88	5.40	12.08	89.78	93.04
	BERT-Attack	—	—	8.86	23.27	12.33	25.48	27.12	37.44	30.40	40.10
	Sep-Attack	81.61	42.87	9.38	22.99	11.28	25.45	26.13	39.24	93.61	95.30
	Co-Attack	81.73	39.26	10.53	23.62	12.54	26.05	27.24	40.62	95.91	96.50
	SGA	83.42	38.64	15.64	28.60	18.02	33.07	39.02	51.45	99.87	99.90
	DRA	82.92	39.56	19.92	35.87	22.02	38.24	45.70	**61.92**	**100.00**	99.93
	Ours	**83.46**	**37.31**	**21.38**	**36.65**	**22.87**	**39.71**	**46.99**	61.40	99.87	**99.93**

The source column presents the VLP models used to generate multimodal adversarial examples. The shaded gray area represents experiments conducted in a white-box attack setting, while the remaining entries correspond to black-box attacks. We report the R@1 ASR (%) performance of each model in both ITR, and visualize the ratio (%) and FID for visual fidelity.

1) Transferability of Adversarial Attacks on Models With Same Architecture: ALBEF and TCL adopt a fusion-based architecture. Although their pretraining objectives differ, their overall network structures share certain similarities. When using ALBEF as the source attack model and TCL as the target model, our method achieves a higher ASR than SGA in both TR R@1 and IR R@1, with TR R@1 improving by 2%-3% and IR R@1 improving by 2.3%-3.1%. Conversely, when TCL is used as the source model and ALBEF as the target model, our method outperforms SGA by approximately 4.9% in TR R@1 and 5.2% in IR R@1, demonstrating stronger transferability in adversarial attacks. This indicates that even between models with similar architectures, an optimized attack strategy can still yield significant improvements.

In contrast, when both the source and target models follow an alignment-based architecture but have significantly different encoder structures, the transferability of attacks becomes more restricted. For example, CLIP-ViT uses a Vision Transformer as its image encoder, whereas CLIP-CNN employs a ResNet-101. In this case, traditional attack methods exhibit relatively low transfer success rates. PGD, BERT-Attack, Sep-Attack, and Co-Attack all fail to achieve satisfactory transferability between these models. However, our method still improves the ASR by 7.97%-12.37% over SGA, indicating that it is effective not only in models with similar architectures but also in cases where the feature encoding strategies differ.

2) Transferability of Adversarial Attacks Across Different Architectures: In real-world applications, an attacker typically cannot ensure that the target model shares the same architecture as the source model. Therefore, the effectiveness of cross-architecture transferability is crucial in assessing the generalization capability of adversarial attacks. Experimental results show that traditional methods perform poorly in cross-architecture attacks. For instance, when ALBEF is used as the source model and CLIP-ViT as the target model, the TR R@1 of PGD and BERT-Attack is only around 10%-30%, while IR R@1 remains below 50%, indicating that conventional attack methods struggle to adapt to models with different encoder structures.

In contrast, our method significantly outperforms SGA in cross-architecture transferability. Specifically, when transferring from ALBEF to CLIP-ViT, our method achieves TR R@1 and IR R@1 of 53.01% and 63.69%, representing improvements of 16.32% and 16.91% over SGA, respectively. Similarly, when transferring from TCL to CLIP-CNN, our method attains TR R@1 and IR R@1 of 59.26% and 68.71%, with improvements of 17.37% and 17.18%, respectively. These results indicate that our method can generate highly effective adversarial examples even in scenarios with significant differences in encoder architectures, demonstrating superior attack transferability.

Overall, our method not only achieves an extremely high ASR in white-box attacks but also significantly outperforms existing methods in both same-architecture and cross-architecture black-box transfer attacks. Particularly in cross-architecture transferability, our method improves ASR by 16%–18% over SGA, further validating its superior generalization capability.

C. Cross-Task Transferability of Adversarial Attacks

To further assess the transferability of adversarial examples, we first generate adversarial samples on the ITR task and

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

TABLE II
COMPARISON WITH MAINSTREAM METHODS IN ITR ON THE MSCOCO DATASET

Source	Attack	Ratio	FID	ALBEF		TCL		CLIP-ViT		CLIP-CNN	
				TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1
ALBEF	PGD	85.24	35.91	94.35	93.26	34.15	36.86	21.71	27.06	23.83	30.96
	BERT-Attack	—	—	11.57	27.46	12.64	28.07	29.33	43.17	32.69	46.11
	Sep-Attack	85.24	35.91	97.01	96.59	61.06	66.13	57.19	65.82	58.81	68.61
	Co-Attack	85.93	35.84	96.65	98.33	57.33	64.19	50.71	57.36	52.06	60.74
	SGA	87.12	**33.66**	**99.95**	99.94	87.46	88.17	63.72	69.71	63.91	70.78
	DRA	86.78	34.57	99.90	99.93	89.47	90.77	66.99	72.78	65.95	74.68
	Ours	**87.41**	33.84	99.92	**99.95**	**90.93**	**90.87**	**68.64**	**73.41**	**68.37**	**75.05**
TCL	PGD	82.55	41.24	40.81	44.09	98.54	98.20	21.79	26.92	24.97	32.17
	BERT-Attack	—	—	11.89	26.82	14.54	29.17	29.69	44.49	33.46	46.07
	Sep-Attack	82.55	41.24	66.38	72.19	99.15	98.98	59.94	65.95	60.77	69.37
	Co-Attack	82.41	40.05	65.22	72.41	94.95	97.87	55.28	62.38	56.68	66.45
	SGA	**84.86**	**37.85**	92.70	92.96	**100.00**	**100.00**	60.29	63.21	60.52	67.34
	DRA	83.94	38.32	93.64	**95.53**	**100.00**	99.99	64.32	**70.99**	64.95	72.09
	Ours	83.67	38.57	**93.68**	93.94	**100.00**	**100.00**	**65.62**	69.67	**66.24**	**72.18**
CLIP-ViT	PGD	79.13	45.82	10.26	13.69	12.72	15.81	82.91	90.51	21.62	28.78
	BERT-Attack	—	—	9.59	22.64	11.80	25.07	28.34	39.08	30.40	37.43
	Sep-Attack	79.13	45.82	25.91	36.84	28.20	38.47	88.36	97.09	47.57	57.79
	Co-Attack	78.58	46.13	26.35	31.66	28.23	38.42	88.78	96.72	47.36	58.45
	SGA	81.70	39.37	43.75	51.08	44.05	51.02	**100.00**	**100.00**	70.66	75.58
	DRA	81.38	39.58	45.38	51.62	45.01	51.47	**100.00**	**100.00**	72.38	**78.53**
	Ours	**82.12**	**38.62**	**46.30**	**53.80**	**46.14**	**52.76**	**100.00**	**100.00**	**73.40**	78.25
CLIP-CNN	PGD	82.22	40.56	8.38	12.73	11.90	15.68	13.66	20.62	92.68	94.71
	BERT-Attack	—	—	8.86	23.27	12.33	25.48	27.12	37.44	30.40	40.10
	Sep-Attack	82.22	40.56	29.41	40.64	31.41	42.99	52.23	59.73	96.16	97.54
	Co-Attack	81.86	40.70	29.49	41.01	31.83	43.44	53.15	61.56	97.79	98.45
	SGA	**83.54**	37.26	36.94	46.79	38.81	48.90	62.19	67.73	**99.92**	99.97
	DRA	83.05	37.57	38.05	47.50	**41.47**	**50.49**	63.11	**68.10**	99.75	99.93
	Ours	83.16	**37.14**	**38.31**	**47.86**	40.11	49.63	**63.87**	67.84	**100.00**	**100.00**

The source column presents the VLP models used to generate multimodal adversarial examples. The shaded gray area represents white-box attacks, while the remaining entries are black-box attacks. We report the R@1 ASR (%) performance of each model in both ITR, and visualize the ratio (%) and fid for visual fidelity.

TABLE III
CROSS-TASK ADVERSARIAL TRANSFERABILITY

Attack	ITR–VG			ITR–IC				
	Val	Test A	Test B	B@4	METEOR	ROUGE-L	CIDEr	SPICE
Baseline	54.2	61.7	43.3	20	23.6	45.9	84.6	18.6
SGA	45.8	50.6	37.5	13.6	18.6	38.5	55.6	13.2
DRA	45.2	**49.9**	36.6	12.8	18.1	37.7	53.3	12.9
Ours	**44.2**	50.2	**36.4**	**12.3**	**17.7**	**37.2**	**51.6**	**12.4**

We use ALBEF to generate multimodal adversarial examples and apply attacks on the visual grounding (VG) task in the RefCOCO+ dataset and the image captioning (IC) task in the MSCOCO dataset. The baseline represents the original performance of each task on clean samples, where a lower value indicates a greater impact of the adversarial attack and higher attack effectiveness.

then evaluate their effectiveness on two additional tasks: visual grounding and image captioning. This allows us to verify their applicability and effectiveness in cross-task scenarios.

1) Attack on IC: In our experiments, we use ALBEF, trained on the ITR task, to generate adversarial images and evaluate their impact on BLIP [37], a state-of-the-art image captioning model selected for its strong performance and wide adoption in previous work. The experiment is conducted on the MSCOCO dataset, and we evaluate the quality of generated captions using BLEU [38], METEOR [39], ROUGE [40], CIDEr [41], and SPICE [42] scores. As shown in Table III, our method suc cessfully degrades text generation quality across all evaluation metrics. Compared to the baseline, BLEU-4 decreases by 7.7%, CIDEr drops by 33.0%, and SPICE decreases by 6.2%, indicating that adversarial samples effectively disrupt the model's text generation ability. Compared to SGA, our method further reduces CIDEr by 4.0% and SPICE by 0.8%, demonstrating stronger attack performance. These results confirm the effectiveness of our approach in cross-task adversarial attacks, leading to more distorted generated text and further validating the transferability of adversarial examples in vision-language tasks. The visualization results of the generated adversarial examples and their effects on image captioning are illustrated in Fig. 2.

2) Attack on VG: We use ALBEF as the source model in the ITR task to generate adversarial images and then attack ALBEF as the target model in the visual grounding task. The results on the RefCOCO+ dataset are shown in Table III. Compared to the baseline, both our method and SGA lead to a performance drop across all test sets, indicating that the adversarial samples successfully weaken the model's visual localization capability. Notably, our method outperforms SGA in Test A and Test B, with R@1 in Test B further decreasing by 1.1%, suggesting that our attack strategy is more effective in influencing the cross-task transferability of the target model. Additionally, our method performs comparably to SGA on the Val while demonstrating overall more stable attack performance, further confirming its effectiveness in vision-based tasks. Fig. 3 shows qualitative examples highlighting the impact of adversarial attacks on the visual grounding task.

Building upon the quantitative metrics and visualization results discussed above, we further explore the intrinsic frequency characteristics of adversarial examples. To this end, we present a comparative visualization of the log-amplitude spectra under three configurations: clean images, adversarial examples without frequency-domain augmentation, and adversarial examples incorporating frequency-domain augmentation. As

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

Fig. 2. Visualization of IC. We use the ALBEF model, built on the ITR task, to generate adversarial images on the MSCOCO dataset. Then, we employ the BLIP model to generate image captions for both the original and adversarial images. Compared to clean images, the captions generated from adversarial examples often exhibit inaccuracies or hallucinations. This discrepancy is also quantitatively reflected in the evaluation metrics (for example, the BLEU-4 score decreases by 7.7% and the CIDEr score decreases by 33.0%, as shown in Table III), indicating a significant degradation in caption quality.

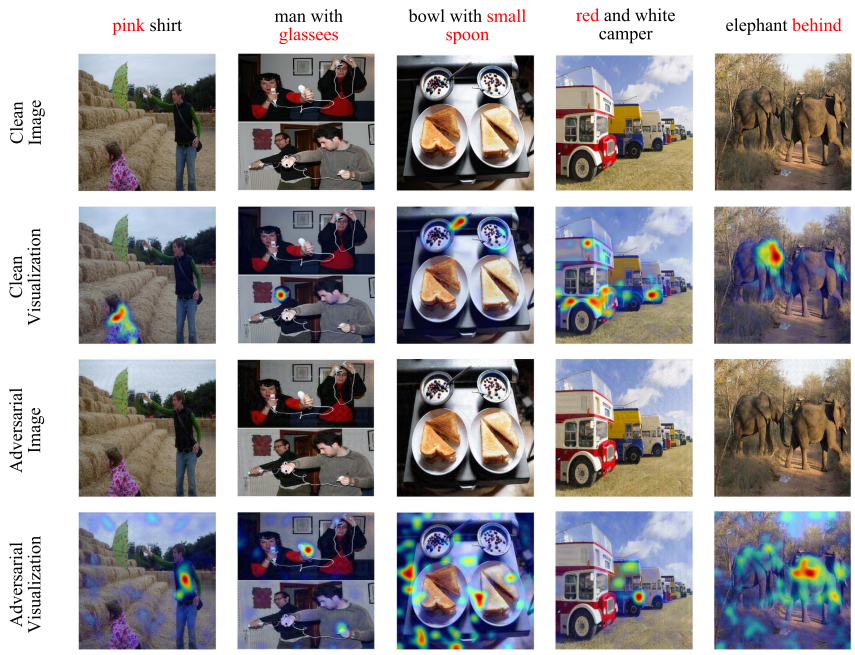


Fig. 3. Visualization of VG. We use the ALBEF model to generate adversarial images on the RefCOCO+ dataset. Then, we apply the same model in the VG task to identify target regions in both the original and adversarial images, locating the areas corresponding to the red-labeled words. Adversarial perturbations cause the predicted attention regions to shift, leading the model to focus on irrelevant areas or fail to locate the correct targets. Table III further demonstrates the performance degradation, validating the impact of the adversarial attacks.

shown in Fig. 4, adversarial perturbations without frequency constraints introduce substantial high-frequency noise, resulting in a more scattered energy distribution in the spectrum. In contrast, frequency-domain augmented examples effectively suppress high-frequency components while preserving low-frequency structures, producing more stable and transferable perturbations. This observation further illustrates how frequency-domain augmentation significantly improves attack performance in cross-model and cross-task settings, as demonstrated in Figs. 2 and 3.

D. Adversarial Attacks on LLMs

We used the Flickr30 K dataset and selected the ALBEF model as a surrogate model to generate adversarial examples under $\epsilon_t = 1$ and $\epsilon_v = 8/255$. Subsequently, we fed these adversarial examples into various LLMs, including BLIP2 [43] (BLIP2-OPT-2.7B) and InstructBLIP [44] (InstructBLIP-Vicuna-7B), both of which were employed with their official pretrained weights without further fine-tuning. The experimental results shown in Table IV indicate that, although ours attack was designed for VLP models, it still interfered with the judgment ability of LLMs to some extent, both in cross-model and cross-task scenarios. Moreover, the attack performance on LLMs was superior to the baseline methods.

E. Ablation Study

1) Method Ablation Study: Our proposed method is built upon SGA as the baseline, introducing two key improvements. First, we leverage a frequency-domain augmentation mechanism to optimize the core features of images while removing

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

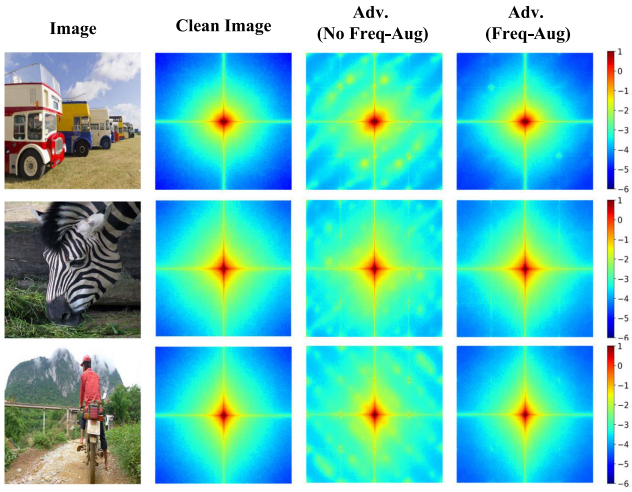


Fig. 4. Visualization comparison of the frequency spectra of clean images and two types of adversarial examples. Frequency-domain augmentation effectively suppresses high-frequency components while preserving low-frequency structures, resulting in more stable and transferable adversarial perturbations.

TABLE IV
TRANSFER ATTACK PERFORMANCE OF LLMS ON THE FLICKR30 K DATASET

Model	Attack	B@4	METEOR	ROUGE-L	CIDEr	SPICE
InstructBLIP	Baseline	47.57	30.98	62.49	122.17	15.64
	Co-attack	25.89	22.23	48.51	67.52	15.33
	SGA	24.42	21.00	46.91	61.72	14.35
	DRA	23.81	20.72	46.65	**60.14**	14.29
	Ours	**23.60**	**20.60**	**46.50**	61.10	**14.20**
BLIP2	Baseline	41.45	27.20	57.96	100.95	13.32
	Co-attack	25.23	20.57	48.29	60.42	14.08
	SGA	24.16	19.47	46.98	55.77	13.27
	DRA	23.60	19.61	46.90	55.58	13.15
	Ours	**22.50**	**19.00**	**46.20**	**53.20**	**12.60**

The adversarial examples generated by ALBEF for the ITR task are transferred to LLMS for evaluation on the IC task. The baseline represents the original performance of each task on clean samples, where a lower value indicates a greater impact of the adversarial attack and higher attack effectiveness.

TABLE V
THE ATTACK PERFORMANCE IN THE ITR TASK UNDER DIFFERENT SETTINGS OF THE ATTACK PARAMETER λ ON THE FLICKR30 K DATASET

Source	Parameter	ALBEF	TCL	CLIP-ViT	CLIP-CNN	Average
ALBEF	$\lambda = 0.1$	99.95	89.92	63.68	63.71	79.32
	$\lambda = 0.3$	99.97	91.67	71.46	73.44	84.14
	$\lambda = 0.5$	99.92	90.93	68.64	68.37	81.97
	$\lambda = 0.7$	**99.95**	**92.04**	**75.39**	**75.19**	**85.64**
	$\lambda = 0.9$	99.92	90.85	72.59	70.19	83.39
TCL	$\lambda = 0.1$	93.25	100.00	62.65	63.06	79.74
	$\lambda = 0.3$	94.43	100.00	70.85	72.25	84.38
	$\lambda = 0.5$	95.20	100.00	74.78	75.64	86.41
	$\lambda = 0.7$	**95.36**	**100.00**	**75.70**	**77.73**	**87.20**
	$\lambda = 0.9$	95.08	100.00	74.93	75.36	86.34
CLIP-ViT	$\lambda = 0.1$	45.32	44.50	99.90	71.72	65.36
	$\lambda = 0.3$	49.39	49.95	99.90	78.46	69.43
	$\lambda = 0.5$	**52.69**	**52.49**	100.00	80.94	**71.53**
	$\lambda = 0.7$	51.80	51.49	**100.00**	**82.51**	71.45
	$\lambda = 0.9$	51.27	51.90	99.90	81.52	71.15
CLIP-CNN	$\lambda = 0.1$	34.00	37.65	58.15	99.75	57.39
	$\lambda = 0.3$	37.54	41.91	67.23	99.80	61.62
	$\lambda = 0.5$	38.31	40.11	63.87	99.80	60.52
	$\lambda = 0.7$	**42.25**	**45.05**	72.72	**100.00**	**65.01**
	$\lambda = 0.9$	39.27	42.89	**72.80**	99.65	63.65

The attack effectiveness is measured using the R@1 metric. The values chosen for the parameter λ were 0.1, 0.3, 0.5, 0.7, and 0.9.

TABLE VI
THE ATTACK PERFORMANCE IN THE ITR TASK UNDER DIFFERENT SETTINGS OF THE ATTACK PARAMETER λ ON THE MSCOCO DATASET

Source	Parameter	ALBEF	TCL	CLIP-ViT	CLIP-CNN	Average
ALBEF	$\lambda = 0.1$	99.94	89.97	70.99	73.14	83.51
	$\lambda = 0.3$	99.93	92.06	78.41	80.07	87.62
	$\lambda = 0.5$	**99.95**	90.87	73.41	75.05	84.82
	$\lambda = 0.7$	99.93	**92.28**	**80.43**	**81.55**	**88.55**
	$\lambda = 0.9$	**99.95**	90.19	75.23	78.68	86.01
TCL	$\lambda = 0.1$	93.77	99.99	69.04	71.33	83.53
	$\lambda = 0.3$	95.22	99.99	77.09	78.79	87.77
	$\lambda = 0.5$	95.76	100.00	80.20	81.27	89.31
	$\lambda = 0.7$	96.38	**100.00**	80.88	**81.99**	**89.81**
	$\lambda = 0.9$	**96.40**	100.00	**81.23**	81.36	89.75
CLIP-ViT	$\lambda = 0.1$	54.03	53.69	99.99	78.45	71.54
	$\lambda = 0.3$	60.55	59.70	100.00	84.37	76.16
	$\lambda = 0.5$	61.65	60.47	100.00	84.92	76.76
	$\lambda = 0.7$	**63.00**	**61.67**	**100.00**	**86.41**	**77.77**
	$\lambda = 0.9$	62.85	61.31	99.80	85.34	77.33
CLIP-CNN	$\lambda = 0.1$	44.85	47.25	65.37	99.97	64.36
	$\lambda = 0.3$	50.43	50.63	71.22	99.93	68.05
	$\lambda = 0.5$	47.86	49.63	67.84	99.95	66.32
	$\lambda = 0.7$	**54.55**	**56.41**	**77.02**	**100.00**	**72.00**
	$\lambda = 0.9$	51.18	53.05	73.23	99.81	69.32

The attack effectiveness is measured using the R@1 metric. The values chosen for the parameter λ were 0.1, 0.3, 0.5, 0.7, and 0.9.

unnecessary high-frequency components, resulting in a more concise and stable representation for generating adversarial images. Second, inspired by the image frequency-domain augmentation mechanism, we propose a multi-level text feature extraction strategy, enabling more fine-grained interaction guidance for generating adversarial text. To analyze the impact of each improvement on the effectiveness of our method, we conduct an ablation study on the ITR task, using the transferability from ALBEF to three other VLP models as the evaluation metric.

In the ablation study, we systematically remove each enhancement from our method and compare their transferability with the baseline SGA and our final method, which incorporates both improvements. The results are shown in Fig. 5. It is evident that in Setting 1, whether attacking TCL, which shares the same architecture as ALBEF, or attacking an alignment-based VLP model, our method exhibits a significant drop in transferability. However, in Setting 2, our method also shows a declining trend, though to a lesser extent. This indicates that the frequency-domain augmentation mechanism plays the most crucial role in our method, while the multi-level text feature extraction strategy also contributes to improving the overall effectiveness of our approach.

2) Parameter Selection: To achieve finer-grained guidance in the generation of adversarial text, we incorporate traditional methods by computing the KL divergence between the original text embeddings and masked text embeddings to measure the impact of individual words on the overall sentence. Additionally, we utilize the multi-level text feature extraction strategy to compute the cosine similarity between word-level and sentence-level features, which quantifies the influence of a word in the overall context. The ablation study in the method settings section has already demonstrated the effectiveness of our multi-level text feature extraction strategy. Therefore, we balance the KL divergence score and the cosine similarity score through a weighted combination. For evaluation, we select a fusion-based model (TCL) and an alignment-based model (CLIP-ViT), assessing the attack transferability under different parameter settings when transferring adversarial examples both within the same model and to three other models. The experimental results are summarized in Tables V and VI. The changing trend is shown in Fig. 6. From the experimental results, it can be observed that the optimal performance is achieved when the parameter is set to 0.7. This further validates that the proposed multi-level text feature extraction strategy is effective in mitigating the overfitting problem.

3) Frequency Range of the Mask Matrix: We generate adversarial examples based on the ALBEF source model and

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

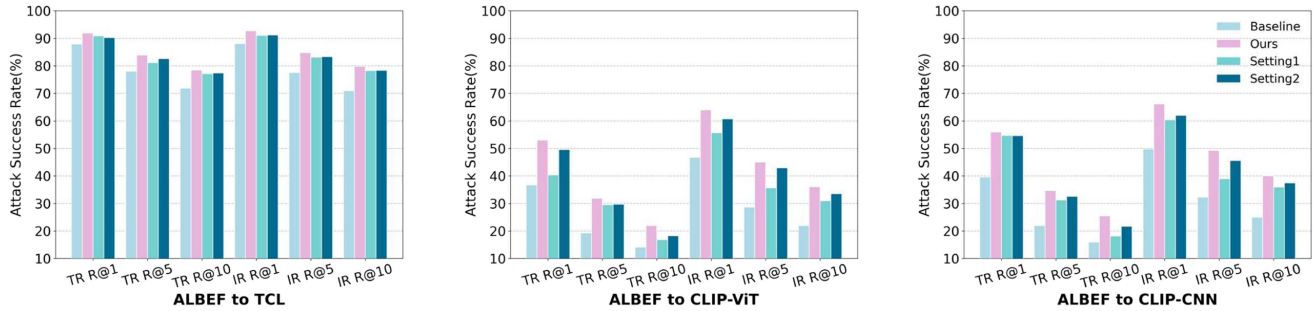


Fig. 5. Method ablation study. The Baseline is set as SGA. Setting 1 removes the frequency-domain enhancement mechanism used to optimize core image features and eliminate unnecessary high-frequency components. Setting 2 removes the multi-level text feature extraction strategy, which provides fine-grained interactive guidance for generating adversarial text.

TABLE VII
COMPARISON OF IMAGE QUALITY AND MODEL PERFORMANCE ACROSS DIFFERENT FREQUENCY MASK RANGES

Mask Range	Ratio	FID	ALBEF		TCL		CLIP-ViT		CLIP-CNN	
			TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1	TR R@1	IR R@1
20	**89.40**	**34.56**	100.00	99.93	91.25	92.45	53.01	63.69	57.22	66.38
40	84.30	37.84	100.00	99.93	91.99	93.05	52.06	63.60	55.56	66.21
60	81.10	42.45	100.00	99.93	92.62	93.38	52.55	63.31	56.07	66.93
80	81.20	42.37	99.90	99.93	92.41	92.93	51.57	63.08	56.32	65.59
100	82.50	40.63	99.90	99.98	91.68	92.74	51.94	62.56	53.77	65.15
120	84.30	38.24	100.00	99.98	91.57	91.74	51.82	62.50	52.62	64.87

The SSIM, PSNR, and FID are adopted as criteria to evaluate the quality of adversarial examples. Ratio (%) represents the proportion of images satisfying both SSIM > 0.8 and PSNR > 30db, with the highest value highlighted in bold.

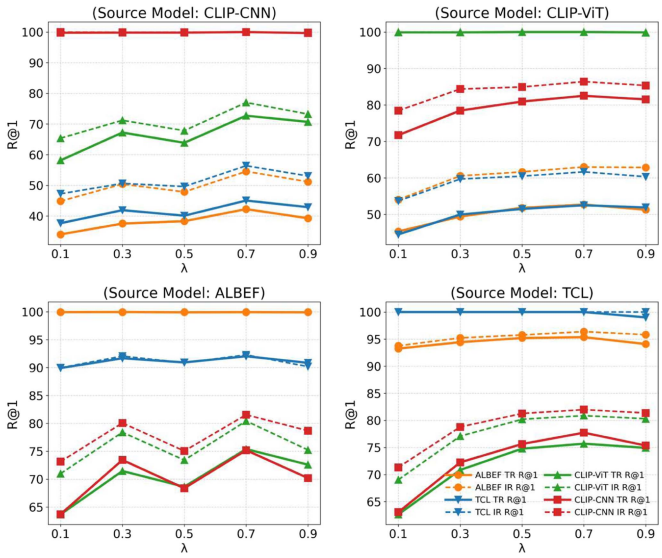


Fig. 6. Parameter selection. Variations in ASR for the ITR task under different parameter settings. The four subplots correspond to different source models: CLIP-CNN, CLIP-ViT, ALBEF, and TCL. The x-axis represents the parameter λ, while the y-axis indicates the R@1 metric. Curves in different colors and markers represent different target models, visually demonstrating the impact of parameter selection on adversarial attack effectiveness across various architectures.

transfer them to four models—ALBEF, TCL, CLIP-ViT, and CLIP-CNN—to evaluate the impact of different mask ranges on ASR. The experiment selects mask ranges of 20, 40, 60, 80, 100, and 120, calculates the ASR for each, and further examines the quality of adversarial examples (measured by the proportion meeting the criteria of SSIM $>$ 0.8 and PSNR $>$ 30 dB), as well as the FID score. The experimental results are shown in Table VII.

Experimental results show that variations in ASR across different mask ranges are relatively small, indicating that mask range adjustments have little impact on attack transferability. For example, TR R@1 and IR R@1 exhibit only slight fluctuations in ALBEF and TCL models and remain relatively stable in CLIP-ViT and CLIP-CNN models. Moreover, while increasing the mask range maintains attack effectiveness, an excessively large mask range may degrade the quality of adversarial examples, as indicated by the decreasing proportion of samples meeting the criteria of SSIM $>$ 0.8 and PSNR $>$ 30 dB, as well as the increasing FID score. Specifically, when the mask range is set to 20, this proportion reaches 89.4% and achieves the lowest FID score, which is the best among all settings, indicating that the adversarial examples generated under this setting have the highest quality. In summary, although the mask range has a limited impact on ASR, selecting a smaller mask range (20) ensures both attack effectiveness and improves adversarial example quality. Therefore, this study ultimately selects a mask range of 20 as the optimal setting to balance attack transferability and sample quality.

V. CONCLUSION

This study proposes a multimodal adversarial attack method based on frequency-domain enhancement and fine-grained cross-modal guidance to improve attack stability and transferability. The frequency-domain mechanism reduces high-frequency noise in images, enhancing cross-model and

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

cross-task transferability, while fine-grained text feature extraction optimizes image-text alignment for better attack effectiveness. Experiments show that our method outperforms existing approaches, increasing black-box ASR by 16%-18% and remaining effective when transferring adversarial examples to large language models. Ablation studies confirm the key role of both frequency enhancement and fine-grained text extraction. This work expands multimodal adversarial attack research and offers insights into vision-language model security. Future research should focus on improving adversarial stability and generalization across more complex tasks and diverse models.

References

[1] H. Chen, G. Ding, X. Liu, Z. Lin, J. Liu, and J. Han, "IMRAM: Iterative matching with recurrent attention memory for cross-modal image-text retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 12652–12660.

[2] M. Cheng et al., "ViSTA: Vision and scene text aggregation for cross-modal retrieval," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 5174–5183.

[3] X. Liu, Z. Wang, J. Shao, X. Wang, and H. Li, "Improving referring expression grounding with cross-modal attention-guided erasing," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 1950–1959.

[4] S. Yang, G. Li, and Y. Yu, "Cross-modal relationship inference for grounding referring expressions," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2019, pp. 4140–4149.

[5] H. Zhang, Y. Niu, and S.-F. Chang, "Grounding referring expressions in images by variational context," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 4158–4166.

[6] K. Xu, "Show, attend and tell: Neural image caption generation with visual attention," 2015, *arXiv:1502.03044*.

[7] B. Wang, L. Ma, W. Zhang, and W. Liu, "Reconstruction network for video captioning," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2018, pp. 7622–7631.

[8] S. Bai and S. An, "A survey on automatic image caption generation," *Neurocomputing*, vol. 311, pp. 291–304, 2018.

[9] C. Zhao et al., "Adversarial training via multi-guidance and historical memory enhancement," *Neurocomputing*, vol. 619, 2025, Art. no. 129124. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S0925231224018952

[10] D. Lu, Z. Wang, T. Wang, W. Guan, H. Gao, and F. Zheng, "Set-level guidance attack: Boosting adversarial transferability of vision-language pre-training models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 102–111.

[11] Y. Qian et al., "F^{2}2AT: Feature-focusing adversarial training via disentanglement of natural and perturbed patterns," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 9, pp. 5201–5213, Sep. 2025.

[12] S. Gao, X. Jia, X. Ren, I. Tsang, and Q. Guo, "Boosting transferability in vision-language attacks via diversification along the intersection region of adversarial trajectory," in *Proc. Eur. Conf. Comput. Vis.*, 2025, pp. 442–460.

[13] Y. Qian, S. He, C. Zhao, J. Sha, W. Wang, and B. Wang, "LEA2: A lightweight ensemble adversarial attack via non-overlapping vulnerable frequency regions," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2023, pp. 4487–4498.

[14] D. Yin, R. G. Lopes, J. Shlens, E. D. Cubuk, and J. Gilmer, "A fourier perspective on model robustness in computer vision," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2019, Art. no. 1189.

[15] H. Wang, X. Wu, Z. Huang, and E. P. Xing, "High-frequency component helps explain the generalization of convolutional neural networks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2020, pp. 8681–8691.

[16] Z. Wang, Y. Yang, A. Shrivastava, V. Rawal, and Z. Ding, "Towards frequency-based explanation for robust CNN," 2020, *arXiv:2005.03141*.

[17] Y. Qian et al., "Enhancing transferability of adversarial examples through mixed-frequency inputs," *IEEE Trans. Inf. Forensics Secur.*, vol. 19, pp. 7633–7645, Jul. 2024.

[18] L. Li, R. Ma, Q. Guo, X. Xue, and X. Qiu, "BERT-Attack: Adversarial attack against bert using BERT," 2020, *arXiv:2004.09984*.

[19] J. Zhang, Q. Yi, and J. Sang, "Towards adversarial attack on vision-language pre-training models," in *Proc. 30th ACM Int. Conf. Multimedia*, 2022, pp. 5005–5013.

[20] Y. Sharma, G. W. Ding, and M. Brubaker, "On the effectiveness of low frequency perturbations," 2019, *arXiv:1903.00073*.

[21] F.-L. Chen et al., "VLP: A survey on vision-language pre-training," *Mach. Intell. Res.*, vol. 20, no. 1, pp. 38–56, 2023.

[22] B. A. Plummer, L. Wang, C. M. Cervantes, J. C. Caicedo, J. Hockenmaier, and S. Lazebnik, "Flickr30k entities: Collecting region-to-phrase correspondences for richer image-to-sentence models," in *Proc. IEEE Int. Conf. Comput. Vis.*, 2015, pp. 2641–2649.

[23] T.-Y. Lin et al., "Microsoft COCO: Common objects in context," in *Proc. 13th Eur. Conf. Comput. Vis.*, Springer, 2014, pp. 740–755.

[24] J. Li, R. Selvaraju, A. Gotmare, S. Joty, C. Xiong, and S. C. H. Hoi, "Align before fuse: Vision and language representation learning with momentum distillation," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2021, pp. 9694–9705.

[25] J. Yang et al., "Vision-language pre-training with triple contrastive learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, 2022, pp. 15671–15680.

[26] A. Radford et al., "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn.*, 2021, pp. 8748–8763.

[27] A. Madry, A. Makelov, L. Schmidt, D. Tsipras, and A. Vladu, "Towards deep learning models resistant to adversarial attacks," in *Proc. Int. Conf. Learn. Representations*, 2018. [Online]. Available: https://openreview.net/forum?id=rJzIBfZAb

[28] H. Wang et al., "Transferable multimodal attack on vision-language pre-training models," in *Proc. 2024 IEEE Symp. Secur. Privacy*, 2024, pp. 1722–1740.

[29] C. Szegedy et al., "Intriguing properties of neural networks," 2013, *arXiv:1312.6199*.

[30] Y. Taigman, M. Yang, M. Ranzato, and L. Wolf, "DeepFace: Closing the gap to human-level performance in face verification," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2014, pp. 1701–1708.

[31] C. Guo, J. S. Frank, and K. Q. Weinberger, "Low frequency adversarial perturbation," 2018, *arXiv:1809.08758*.

[32] W. Zhou et al., "Transferable adversarial perturbations," in *Proc. Eur. Conf. Comput. Vis.*, 2018, pp. 452–467.

[33] L. Zhang, Y. Luo, H. Shen, and T. Wang, "A fourier perspective of feature extraction and adversarial robustness," in *Proc. 33rd Int. Joint Conf. Artif. Intell.*, 2024, Art. no. 1189.

[34] G. Kim, J. Kim, and J.-S. Lee, "Exploring adversarial robustness of vision transformers in the spectral perspective," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis.*, 2024, pp. 3976–3985.

[35] M. V. Koroteev, "BERT: A review of applications in natural language processing and understanding," 2021, *arXiv:2103.11943*.

[36] L. Yu, P. Poirson, S. Yang, A. C. Berg, and T. L. Berg, "Modeling context in referring expressions," in *Proc. 14th Eur. Conf. Comput. Vis.*, Springer, 2016, pp. 69–85.

[37] J. Li, D. Li, C. Xiong, and S. C. H. Hoi, "BLIP: Bootstrapping language-image pre-training for unified vision-language understanding and generation," in *Proc. Int. Conf. Mach. Learn.*, 2022, pp. 12888–12900.

[38] K. Papineni, S. Roukos, T. Ward, and W.-J. Zhu, "BLEU: A method for automatic evaluation of machine translation," in *Proc. 40th Annu. Meeting Assoc. Comput. Linguistics*, 2002, pp. 311–318.

[39] S. Banerjee and A. Lavie, "METEOR: An automatic metric for MT evaluation with improved correlation with human judgments," in *Proc. ACL Workshop Intrinsic Extrinsic Eval. Measures Mach. Transl. Summarization*, 2005, pp. 65–72.

[40] L. Chin-Yew, "ROUGE: A package for automatic evaluation of summaries," in *Proc. Workshop Text Summarization Branches Out*, 2004, pp. 74–81.

[41] R. Vedantam, C. L. Zitnick, and D. Parikh, "CIDEr: Consensus-based image description evaluation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit.*, 2015, pp. 4566–4575.

[42] P. Anderson, B. Fernando, M. Johnson, and S. Gould, "SPICE: Semantic propositional image caption evaluation," in *Proc. 14th Eur. Conf. Comput. Vis.*, Springer, 2016, pp. 382–398.

[43] J. Li, D. Li, S. Savarese, and S. C. H. Hoi, "BLIP-2: Bootstrapping language-image pre-training with frozen image encoders and large language models," in *Proc. Int. Conf. Mach. Learn.*, 2023, pp. 19730–19742.

[44] W. Dai et al., "InstructBLIP: Towards general-purpose vision-language models with instruction tuning," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2023, Art. no. 2142.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.

Yaguan Qian received the PhD degree in computer science from Zhejiang University, in 2014. He is currently a full professor with the School of Artificial Intelligence and Information Engineering, Zhejiang University of Science and Technology. He has authored or co-authored more than 40 papers in peer-reviewed international conferences and journals including ICCV, ECCV, AAAI, *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Neural Networks and Learning Systems*, *ACM Transactions on Privacy and Security*, etc. His main research interests include AI security, machine learning, pattern recognition, and machine vision.

Qinqin Yu received the BSc degree from the School of Mathematics and Statistics, Hunan University of Technology and Business, in 2021. She is currently working toward the MS degree with the School of Artificial Intelligence and Information Engineering, Zhejiang University of Science and Technology. Her research interests include deep learning, machine learning security, and computer vision.

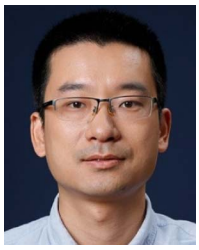
Qiqi Bao received the PhD degree from the Department of Electronic Engineering, Tsinghua University, in 2024. She is currently an assistant researcher with the School of Artificial Intelligence and Information Engineering, Zhejiang University of Science and Technology. She has authored or co-authored 15 papers in peer-reviewed international conferences and journals, including NeurIPS, AAAI, ACM MM, *IEEE Transactions on Image Processing*, *IEEE Transactions on Multimedia*, etc. Her main research interests include image restoration, image generation, and trustworthy artificial intelligence.

Shouling Ji (Member, IEEE) received the PhD degree in electrical and computer engineering from the Georgia Institute of Technology, and the PhD degree in computer science from Georgia State University. He is a full professor with the College of Computer Science and Technology, Zhejiang University and a research faculty with the School of Electrical and Computer Engineering, Georgia Institute of Technology (Georgia Tech). His current research interests include data-driven security and privacy, AI security, and Big Data analytics.

Wei Wang (Member, IEEE) received the PhD degree from Xi'an Jiaotong University, in 2006. He is a full professor with the Ministry of Education Key Lab for Intelligent Networks and Network Security, aka MOE KLINNS Lab, Xi'an Jiaotong University, China. He has authored or co-authored more than 100 peer-reviewed articles in various journals and international conferences. His recent research interests lie in social networks and data security. He is an Elsevier 'highly cited Chinese Researchers'. He is an associate editor of *IEEE Transactions on Dependable and Secure Computing*, an editorial board member of the *Computers & Security*, and a Young AE of the *Frontiers of Computer Science*.

Bin Wang received the PhD degree from the China National Digital Switching System Engineering and Technological R&D Center, in 2010. He is a professor with the Zhejiang Key Laboratory of Artificial Intelligence of Things (AIoT) Network and Data Security, Application and Cybersecurity and Zhejiang University. His research interests mainly include Internet of Things security, cryptography, artificial intelligence security, new network security architecture and so on.

Zhaoquan Gu (Member, IEEE) received the bachelor's degree in computer science from Tsinghua University, in 2011, and the PhD degree in computer science from Tsinghua University, in 2015. He is currently a professor with the School of Computer Science and Technology, Harbin Institute of Technology (Shenzhen), China. He is also a professor with the Department of New Networks, Peng Cheng Laboratory, Shenzhen, China. His research interests include cyberspace security, cyber range, Big Data analysis, and artificial intelligence security.



Zhen Lei (Fellow, IEEE) received the BS degree in automation from the University of Science and Technology of China, in 2005, and the PhD degree from the Institute of Automation, Chinese Academy of Sciences, in 2010, where he is currently a professor. He is an IAPR/AAIA fellow. He has published more than 200 papers in international journals and conferences with 34000+ citations in Google Scholar and h-index 87. He was the program co-chair of IJCB2023, was competition co-chair of IJCB2022 and has served as area chairs for several conferences and is associate editor of *IEEE Transactions on Information Forensics and Security*, *IEEE Transactions on Biometrics, Behavior, and Identity Science*, *Pattern Recognition*, *Neurocomputing*, and *IET Computer Vision* journals. His research interests include computer vision, pattern recognition, image processing, and face recognition in particular. He is the winner of the 2019 IAPR Young Biometrics Investigator Award.

Authorized licensed use limited to: Southeast University. Downloaded on December 09,2025 at 01:40:10 UTC from IEEE Xplore. Restrictions apply.